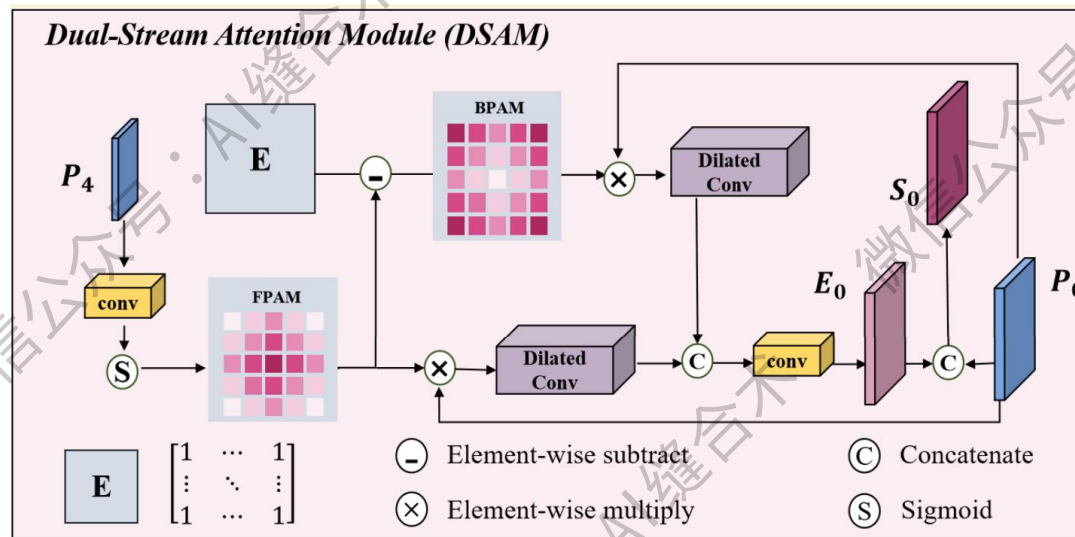


核心速览



论文题目: Boundary-Aware Feature Fusion With Dual-Stream Attention for Remote Sensing Small Object Detection

中文题目: 具有双流注意力的边界感知特征融合用于遥感小目标检测

论文链接: <https://ieeexplore.ieee.org/abstract/document/10787034>

官方 github: <https://github.com/ooo1128/BAFNet>

所属单位: 重庆大学等

核心速览: 本文提出了一种名为边界感知特征融合网络 (BAFNet) 的新框架, 该框架通过双流注意力机制和边界感知监督策略, 显著提升了模型在复杂遥感场景中精确检测小对象的能力。

双流注意力机制 (DSAM)： 利用高层语义特征生成前景和背景注意力图，然后与低层空间特征融合，捕获更具有区分性的前景和背景线索。该机制通过互补学习的方式，有效捕获小物体的显著特征，同时抑制复杂背景的干扰。

```
(b1_conv): ASPP(
  (aspp1): Sequential(
    (0): Conv2d(256, 256, kernel_size=(1, 1), stride=(1, 1))
    (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
  )
  (aspp2): Sequential(
    (0): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(3, 3), dilation=(3, 3))
    (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
  )
  (aspp3): Sequential(
    (0): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(5, 5), dilation=(5, 5))
    (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
  )
  (aspp4): Sequential(
    (0): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(7, 7), dilation=(7, 7))
    (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
  )
)
(rgbd_pred_layer): Pred_Layer(
  (enlayer): Sequential(
    (0): Conv2d(2048, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (2): ReLU(inplace=True)
  )
  (outlayer): Sequential(
    (0): Conv2d(256, 1, kernel_size=(1, 1), stride=(1, 1))
  )
)
)
```

微信公众号：AI缝合术！

```
输入1 feat      : torch.Size([4, 256, 64, 64])
输入2 pred      : torch.Size([4, 1, 64, 64])
输出1 enhanced_feat : torch.Size([4, 256, 64, 64])
输出2 new_pred    : torch.Size([4, 1, 64, 64])
```